# MANASVI SAXENA

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## RESEARCH INTERESTS

Formal Semantics, Static and Dynamic Analysis, Program Verification, Runtime Monitoring

## EDUCATION

**Ph.D. Computer Science, University of Illinois – Urbana Champaign** Dec 2024

- Advisor - Prof. Grigore Roşu
- Thesis - A Semantics-First Approach to Safe Guidelines-Based Clinical Decision Support

**M.S. Computer Science, University of Illinois – Urbana Champaign** May 2018

- Relevant Coursework - Formal Methods of Software Development, Runtime Verification, Logic in Computer Science, Computer and Cryptocurrency Security

**B.S. Computer Science, University of Illinois – Urbana Champaign** May 2015

- Relevant Coursework - Programming Languages and Compilers, Fundamental Algorithms, Theory of Computation, Computer Networking, Artificial Intelligence

## EXPERIENCE

**Formal Systems Laboratory, University of Illinois** May 2016 - December 2024
*Research Assistant* *Urbana, IL*

- Developed the MediK Framework, a DSL for encoding medical guidelines using state machines with strong correctness guarantees (`https://github.com/medik-framework`).
- Maintained RV-Monitor, a language independent monitoring framework that supports rich formalisms for expressing properties.
- Contributed to KEVM, formalizing the semantics of the Ethereum Virtual Machine (EVM) in the K Framework (`http://github.com/kframework/evm-semantics`).

**S3 Automated Reasoning Group, Amazon Web Services (AWS)** Summer 2021 & 2021
*Applied Scientist Intern*

- Developed techniques and tooling for evaluating compliance of highly concurrent, distributed code against relevant properties expressed as RV-Monitor specifications (`https://github.com/runtimeverification/rv-monitor`) utilizing the existing test suite.
- Optimized RV-Monitor, an academic tool, to handle millions of lines of code, and enabled support for expressing properties relevant to concurrent systems, such as read after write consistency and serializability efficiently, enabling RV-Monitor to handle the scale of massively parallel and highly distributed systems like S3.

**Runtime Verification Inc., Urbana, Illinois** July 2015 - May 2016
*Formal Verification Engineer* *Urbana, IL*

- Worked on a team developing a dynamic analysis tool for finding undefined behavior in C programs, using formal semantics of C.
- Worked on improving tool's performance, and evaluated its applications on large codebases.

**Formal Systems Laboratory, University of Illinois** August 2014 - May 2015
*Undergraduate Research Assistant* *Urbana, IL*

- Developed a Semantics-based language-independent Debugger for the K Semantics Framework that allows users to efficiently step through program executions, examine the state space, configuration and rules, among other features.

**Coordinated Science Lab, University of Illinois** Summer 2014
*Undergraduate Research Assistant* *Urbana, IL*

- Worked on visualization of a large dataset containing records of faulty medical devices by the Food and Drug Administration (FDA)

## TEACHING EXPERIENCE

**Teaching Assistant, CS 427** — Fall 2020
*University of Illinois* — *Urbana, IL*

- Helped prepare course materials, and assisted instructor during lectures when needed.
- Held weekly office hours

## PROJECTS

**The MediK Framework** — github.com/medik-framework

- Lead the design and development of the semantics of the MediK language for expressing computer interpretable medical guidelines.

**RV-Monitor** — github.com/runtimeverification/rv-monitor

- Part of core team that maintains and develops RV-Monitor - a language agnostic monitor framework.

**KEVM - Semantics of the EVM in K** — github.com/kframework/evm-semantics

- Worked on formalizing the EVM in K. Worked on capturing the semantics of the Ethereum ABI, as an extension to the original semantics.

## PUBLICATIONS

- Manasvi Saxena. *A semantics-first approach to safe guidelines-based clinical decision support.* PhD thesis, University of Illinois Urbana-Champaign, December 2024. Available at `https://hdl.handle.net/2142/127206`
- Manasvi Saxena, Shuang Song, and Lui Sha. MediK: Towards Safe Guideline-based Clinical Decision Support. In *2023 Formal Methods in Computer-Aided Design (FMCAD)*, pages 306–317. IEEE, 2023
- Shuang Song, Manasvi Saxena, Pei-Hsuan Tsai, and Lui Sha. Towards modular and formally-verifiable software architecture for clinical guidance systems. In *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pages 4271–4276. IEEE, 2023
- Everett Hildenbrandt, Manasvi Saxena, Nishant Rodrigues, Xiaoran Zhu, Philip Daian, Dwight Guth, Brandon Moore, Daejun Park, Yi Zhang, Andrei Stefanescu, et al. KEVM: A complete formal semantics of the Ethereum Virtual Machine. In *2018 IEEE 31st Computer Security Foundations Symposium (CSF)*, pages 204–217. IEEE, 2018
- Daejun Park, Yi Zhang, Manasvi Saxena, Philip Daian, and Grigore Rosu. A formal verification tool for ethereum VM bytecode. In Gary T. Leavens, Alessandro Garcia, and Corina S. Pasareanu, editors, *Proceedings of the 2018 ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/SIGSOFT FSE 2018, Lake Buena Vista, FL, USA, November 04-09, 2018*, pages 912–915. ACM, 2018
- Manasvi Saxena, Nishant Rodrigues, Xiaohong Chen, and Grigore Rosu. Formal Semantics of Hybrid Automata. `http://hdl.handle.net/2142/97207`
- Manasvi Saxena, Xiahong Chen, Shuang Song, Shaoyu Meng, Lui Sha, and Grigore Rosu. Rewriting-Based Computer-Interpretable Clinical Practice Guidelines. `https://hdl.handle.net/2142/116016`
- Dwight Guth, Chris Hathhorn, Manasvi Saxena, and Grigore Rosu. Rv-match: Practical semantics-based program analysis. In *Computer Aided Verification - 28th International Conference, CAV 2016, Toronto, ON, Canada, July 17-23, 2016, Proceedings, Part I*, volume 9779 of *LNCS*, pages 447–453. Springer, July 2016
- Philip Daian, Dwight Guth, Chris Hathhorn, Yilong Li, Edgar Pek, Manasvi Saxena, Traian Florin Serbanuta, and Grigore Rosu. Runtime verification at work: A tutorial. In *Runtime Verification - 16th International Conference, RV 2016 Madrid, Spain, September 23-30, 2016, Proceedings*, volume 10012 of *Lecture Notes in Computer Science*, pages 46–67. Springer, September 2016